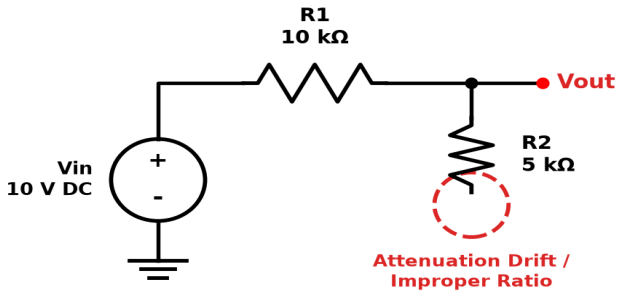
1. Overview

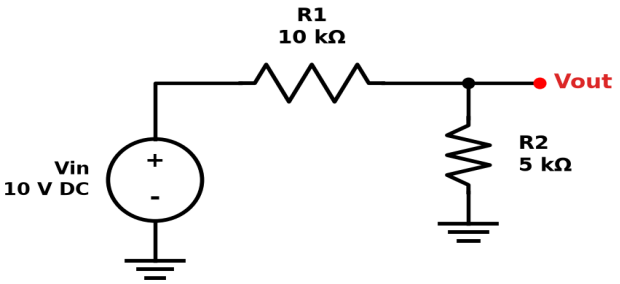
Circuit Type:	Voltage Divider	**Input File:**	voltage_divider_fault.png
Analysis Status:	⚠ 1 Potential Issue Found	**Source:**	Image Upload

2. Circuit Comparison

Uploaded Circuit (Input)



Expected / Reference Circuit



3. Detected Issues

#	Severity	Component / Location	Finding	Recommended Action
1	Error	Branch Net / Node	Topology or parameter mismatch detected.	Inspect component connections and tolerances.

4. Extracted Components

Reference	Component Type	Extracted Value	Unit	Status	Notes
R1	Resistor	10	kΩ	Verified	Value clearly identified
R2	Resistor	5	kΩ	Verified	Value clearly identified
Vin	DC Voltage Source	10	V	Verified	Input source identified

5. Expected Behaviour (Theory)

For a voltage divider, the output relation is given by:

$$V_{out} = V_{in} \cdot \frac{R_2}{R_1 + R_2}$$

Using extracted values:

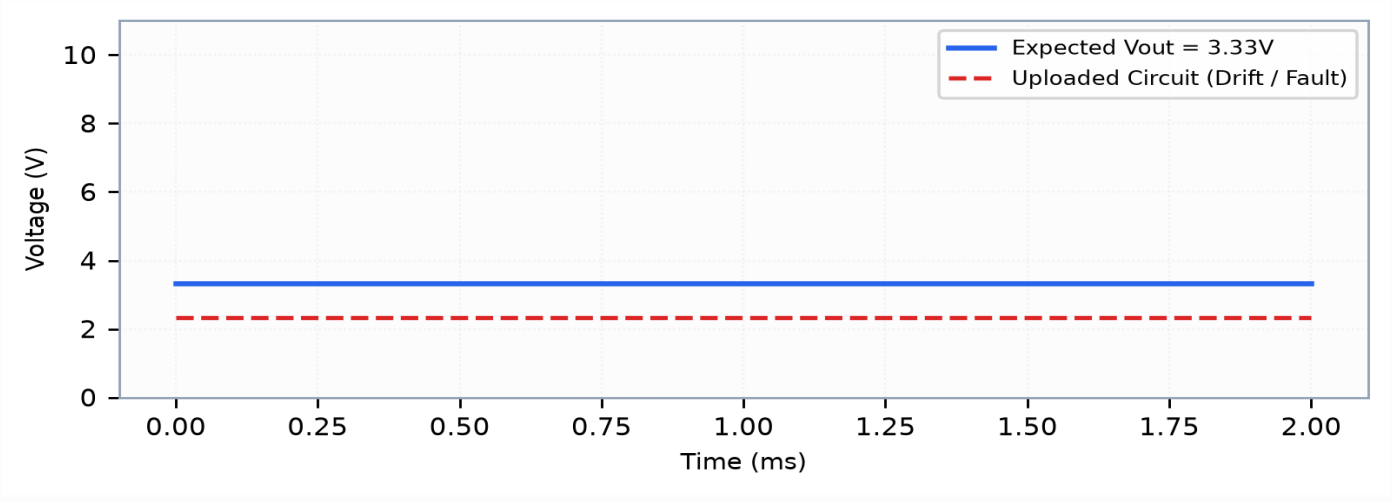
$R_1 = 10\ \text{k}\Omega,\ R_2 = 5\ \text{k}\Omega$

$V_{in} = 10\ \text{V DC}$

$V_{out} = 10 \cdot \frac{5,000}{10,000 + 5,000} \approx 3.33\ \text{V}$

6. Simulation Results

Frequency Response (Magnitude)



7. Recommended Actions

1. Inspect flagged component node and re-solder floating contacts.
2. Verify measured in-circuit component values with DMM meter.
3. Re-run simulation to confirm agreement with nominal reference curve.
4. Check power supply rail decoupling and common ground return path.

8. Conclusion

The uploaded schematic represents an intended voltage divider, but potential connectivity or value shifts prevent nominal electrical behaviour. Resolving the identified findings will align circuit operation with the reference target.